Certified Log-Concavity of the Riemann–Jacobi Kernel and a Source Audit of a Pólya-Type Real-Zero Criterion

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Abstract

We certify strict log-concavity of the Riemann–Jacobi kernel $\Phi(u)$ appearing in the Fourier cosine representation

$$\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du$$

of the Riemann Xi function. The proof combines an elementary algebraic analysis of the dominant term, rigorous interval arithmetic on $[0, 1]$, a no-cancellation certification of $(\log \Phi)'' < 0$ on $[1, 3]$, and an explicit monotonicity tail bound for $u \geq 3$.

We also audit the commonly cited Pólya-type bridge from log-concavity of a kernel to real-rootedness of its Fourier transform. Direct inspection of Pólya (1927) Satz I and Satz II shows that these theorems are, respectively, a preservation theorem and a Mellin–Fourier theorem; neither states the log-concavity criterion needed to derive the Riemann Hypothesis from the result proved here. Accordingly, **no unconditional RH claim is made.** This source audit is part of the contribution: it separates the certified log-concavity theorem from an often-assumed but unverified real-rootedness criterion.

The unconditional contribution of this paper is the certified strict log-concavity of Φ on $[0, \infty)$. The possible implication to RH is recorded only as a conditional consequence of an unsourced Pólya-type criterion (Hypothetical Criterion 13). The proof of log-concavity is independent of the Pólya-source audit.

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1 Introduction

The Riemann Hypothesis (RH), posed by Bernhard Riemann in 1859, asserts that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. Equivalently, all zeros of the Riemann Xi function

$$\Xi(t) := \xi\left(\tfrac{1}{2} + it\right), \quad \xi(s) := \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \tag{1}$$

are real. It is well known [15] that $\Xi(t)$ admits the Fourier cosine representation

$$\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du, \tag{2}$$

where the *Riemann–Jacobi kernel* is

$$\Phi(u) = 4\sum_{n=1}^\infty \varphi_n(u), \quad \varphi_n(u) = \left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\right)e^{-\pi n^2 e^{2u}}. \tag{3}$$

We verify computationally that $\int_0^\infty \Phi(u)\,du = \xi(1/2) \approx 0.4971$ to 15 digits.

Approach. This paper proves a structural property of the Riemann–Jacobi kernel: strict log-concavity of Φ on $[0, \infty)$ (Theorem 12). Because the Fourier transform of this kernel is 2Ξ, this property is naturally relevant to the real-zero problem for Ξ. However, the implication from kernel log-concavity to real-rootedness of the Fourier transform is not established by the Pólya sources audited here. We therefore separate the unconditional log-concavity theorem from the conditional RH implication.

The paper should be read as a certification of log-concavity, not as a proof of RH. The RH consequence appears only in a conditional remark (Remark 16), dependent on a hypothetical criterion not verified in the primary source trail. The source audit (Section 9) is not needed for the proof of Theorem 12; it only governs the interpretation of possible RH consequences.

A second contribution of the paper is therefore source-critical: we trace the commonly cited log-concavity-to-real-rootedness bridge through Pólya (1927), Csordas–Varga (1989), de Bruijn (1950), Ilieff (1955), and later literature, and find no verified primary-source statement matching Hypothetical Criterion 13. In this paper the criterion is consequently treated as an open or unsourced dependency, not as an established theorem.

Prior work. The log-concavity of the dominant term φ_1 was observed by Csordas and Varga [6]. The Turán inequalities for the Taylor coefficients of Ξ were established by Csordas, Norfolk, and Varga [5], and higher-order generalizations by Dimitrov and Lucas [8] and Griffin, Ono, Rolen, and Zagier [10]. The connection between log-concavity and the de Bruijn–Newman constant was explored by de Bruijn [7] and Rodgers and Tao [14]. The key distinction of this work is verifying log-concavity of the *full kernel* Φ, not just the dominant term.

Claims intentionally not made. This paper does not claim:

1. that RH is proved;
2. that Pólya (1927) Satz I or Satz II implies the log-concavity criterion;
3. that Csordas–Varga (1989) Theorem 2.2 states the log-concavity criterion;
4. that Lean verifies RH;
5. that the interval arithmetic certificates are formalized in Lean;
6. that Hypothetical Criterion 13 is established in the peer-reviewed literature.

2 Properties of Φ

Proposition 1. *The kernel Φ defined by* (3) *satisfies:*

(i) $\Phi(u) > 0$ *for all* $u \geq 0$*;*

(ii) $\Phi(-u) = \Phi(u)$ *for all* u*;*

(iii) $\Phi \in L^1(\mathbb{R})$*;*

(iv) $\Phi(u) = O(e^{-\pi e^{2u}})$ *as* $u \to +\infty$*;*

(v) Φ *is real analytic on* $\mathbb{R}$*.*

Proof. Properties (i)–(iii) are classical; see [15, §2.10] and [6, Theorem A]. Property (iv) follows from the dominant exponential factor $e^{-\pi e^{2u}}$ in φ_1. For (v): fix any $u_0 \in \mathbb{R}$. For $r = \pi/8$, any z with $|z - u_0| < r$ satisfies $\mathrm{Re}(e^{2z}) \geq e^{2(u_0 - r)}/\sqrt{2} =: c > 0$. Thus $|\varphi_n(z)| \leq A\, n^4\, e^{-\pi c n^2/2}$ uniformly on $|z - u_0| \leq r/2$. Since $\sum n^4 e^{-\pi c n^2/2} < \infty$, the Weierstrass M-test gives uniform convergence; hence Φ is real analytic at u_0. (Φ is *not* entire; only real analyticity on $\mathbb{R}$, as required by condition (v) of Hypothetical Criterion 13, is claimed.)

Numerical verification. $\Phi(u) > 0$ at 10,001 uniformly spaced points on $[0, 1]$ (min $= 5.51 \times 10^{-7}$ at $u = 1$); $|\Phi(u) - \Phi(-u)|/|\Phi(u)| < 10^{-70}$ at 8 test points. $\square$

3 Algebraic Core

Theorem 2 (Algebraic Core)**.** $(\log \varphi_1)''(u) < 0$ *for all* $u \geq 0$.

Proof. Write $\varphi_1 = g \cdot E$ where $g(u) = \pi e^{5u/2} h(u)$, $h(u) = 2\pi e^{2u} - 3$, and $E(u) = e^{-\pi e^{2u}}$. Since $h(u) \geq 2\pi - 3 > 0$ for $u \geq 0$, we have $\varphi_1 > 0$. Then

$$\log \varphi_1 = \log \pi + \tfrac{5}{2} u + \log h(u) - \pi e^{2u}, \quad (\log \varphi_1)'' = (\log h)'' - 4\pi e^{2u}.$$

With $h' = 4\pi e^{2u}$ and $h'' = 8\pi e^{2u}$:

$$(\log h)'' = \frac{h'' h - (h')^2}{h^2} = \frac{-24\pi e^{2u}}{h(u)^2} < 0.$$

Both terms are negative, so $(\log \varphi_1)'' < 0$. $\square$

Remark 3. $(\log \varphi_1)''(0) \approx -19.56$, monotonically decreasing. The series for $\Phi''(0)$ converges to $-68.052\ldots$ by $N = 5$ terms.

4 Tail Estimate

Lemma 4. *For* $n \geq 2$ *and* $u \geq 0$*:* $e^{-\pi n^2 e^{2u}} \leq e^{-3\pi} \cdot e^{-\pi e^{2u}}$*.*

Proof. Equivalent to $(n^2 - 1)e^{2u} \geq 3$, holding since $n^2 - 1 \geq 3$ and $e^{2u} \geq 1$. $\square$

Proposition 5. *For* $u \geq 0$*:* $|R(u)|/\varphi_1(u) < 1/50$*, where* $R = \sum_{n \geq 2} \varphi_n$*.*

Proof. $|\varphi_n|/\varphi_1 \leq 2n^4 e^{-\pi(n^2-1)e^{2u}}$ (since $B_n(u)/n^4 < 2$). At $u = 0$: $\sum_{n \geq 2} 2n^4 e^{-\pi(n^2-1)} < 0.006 < 1/50$. $\square$

Remark 6. The interval arithmetic retains $n = 1, \ldots, 5$ and omits $\delta := 4 \sum_{n \geq 6} \varphi_n$. Truncation errors on $[0, 1]$: $|\delta| \leq 7.03 \times 10^{-43}$, $|\delta'| \leq 1.18 \times 10^{-39}$, $|\delta''| \leq 1.98 \times 10^{-36}$, certified by interval arithmetic. At $u = 1$, the dominant cross term $|\delta'' \cdot \Phi_5| \approx 1.09 \times 10^{-42}$ gives total error $\leq 1.15 \times 10^{-42}$. Since $|Q_{\Phi_5}| \geq 3.36 \times 10^{-12}$, the safety factor exceeds 2.9×10^{30}.

5 Interval Arithmetic Verification

Definition 7. The *log-concavity numerator* of a positive function f is $Q_f(u) := f''(u) f(u) - f'(u)^2$.

Theorem 8. $Q_\Phi(u) < 0$ *for all* $u \in [0, 1]$.

Proof. Closed-form derivatives:

$$g_n' = 9\pi^2 n^4 e^{9u/2} - \tfrac{15}{2}\pi n^2 e^{5u/2}, \qquad g_n'' = \tfrac{81}{2}\pi^2 n^4 e^{9u/2} - \tfrac{75}{4}\pi n^2 e^{5u/2},$$
$$E_n' = -2\pi n^2 e^{2u} \cdot E_n, \qquad E_n'' = \left(-4\pi n^2 e^{2u} + 4\pi^2 n^4 e^{4u}\right) \cdot E_n.$$

The primary certificate is `proof/verify_logconcavity_arb.py`, using Arb/FLINT ball arithmetic. The independent cross-check is `proof/verify_logconcavity_rigorous.py`, using `mpmath.iv` interval arithmetic at 60-digit precision. Both retain $n = 1, \ldots, 5$ and use the same closed-form derivative formulas above.

The `mpmath.iv` cross-check partitions $[0, 0.949]$ into 1,898 subintervals and $[0.949, 1.0]$ into 51,000 subintervals. The Arb/FLINT verifier uses its own interval partition and precision settings, reported in `results/verify_logconcavity_arb.json`. On every interval, the verifier evaluates Q_{Φ_5} by interval or ball arithmetic and certifies that the upper bound is negative.

Result: The Arb/FLINT primary verifier certifies $Q_{\Phi_5}(u) < 0$ on $[0, 1]$, and the `mpmath.iv` cross-check reproduces the sign certification on all 52,898 subintervals, with maximum `mpmath.iv` upper bound -3.36×10^{-12}. Since truncation error $\le 1.15 \times 10^{-42}$ (Remark 6), $Q_\Phi(u) < 0$ on $[0, 1]$. □

6 Extended Certification on $[1.0, 3.0]$

Direct interval arithmetic on Q_Φ fails for $u > 1$ due to catastrophic cancellation ($40\times$). We certify $(\log \Phi)''(u) < 0$ directly.

Lemma 9 (Uniform tail bound on $[1, 3]$). *For all $u \in [1, 3]$, $|W_{\text{tail}}(u)| \le B(u)$, where $W_{\text{tail}}(u) := (\log \Phi)''(u) - (\log \varphi_1)''(u)$ and $B(u)$ is the interval-arithmetic upper bound defined in Appendix E. The verifier certifies $B(u) \le 1.12 \times 10^{-23}$ on the full cover of $[1, 3]$. The sign proof uses* $\operatorname{upper}(W_1(I_i) + B(I_i)) < 0$ *on each interval; the scalar bound $B \le 1.12 \times 10^{-23}$ is a diagnostic summary.*

Complete derivation in Appendix E.

Theorem 10. $(\log \Phi)''(u) < 0$ *for all* $u \in [1.0, 3.0]$.

All proof-critical estimates used in this certification are derived in Appendix E; the scripts are reproducibility checks and certify numerical constants, not substitutes for the written proof.

Proof. Decompose $(\log \Phi)'' = W_1 + W_{\text{tail}}$ where $W_1(u) = -24\pi e^{2u}/h(u)^2 - 4\pi e^{2u}$ is proved negative in Theorem 2. For each $i = 0, \ldots, 100$, the verifier (`proof/verify_ia_1_to_3_arb.py`, Appendix E) certifies by Arb/FLINT ball arithmetic that $\operatorname{upper}(W_1(I_i) + B(I_i)) < 0$ over $I_i = [u_i - 0.01,\ u_i + 0.01]$, where $B(u)$ is the interval-arithmetic upper bound on $|W_{\text{tail}}(u)|$ certified in Appendix E, Lemma 21. Adjacent intervals overlap; their union $[0.99, 3.01] \supset [1.0, 3.0]$.

Result: All 101 checks certified; maximum $B \le 1.12 \times 10^{-23}$; minimum margin $(-W_1 - B) = 91.3$ at $u = 1.0$. Since $|W_{\text{tail}}| \le B$: $(\log \Phi)'' = W_1 + W_{\text{tail}} \le W_1 + B < 0$. □

7 Tail Bound for $u \ge 3.0$

Theorem 11. $(\log \Phi)''(u) < 0$ *for all* $u \ge 3.0$.

Proof. For $u \ge 3$, Appendix E.6 (Lemmas 23–28) proves

$$|W_{\text{tail}}(u)| \le C_{\text{tail}}\, e^{4u} \varepsilon^*(u),$$

where $C_{\text{tail}} = 2600$. This is entirely separate from and independent of the interval-certified $B(u)$ bound used on $[1, 3]$.

At $u = 3$,

$$W_1(3) < -4\pi e^6 < -1000, \qquad C_{\text{tail}}\, e^{12}\, \varepsilon^*(3) \le 2600\, e^{12} \cdot 32\, e^{-3\pi e^6} < 10^{-1629} < 1.$$

Moreover, $W_1(u)$ is strictly decreasing on $[3, \infty)$ (Lemma 27) and $e^{4u}\varepsilon^*(u)$ is strictly decreasing on $[3, \infty)$ (Lemma 28). Therefore

$$W_1(u) + C_{\text{tail}}\, e^{4u}\varepsilon^*(u) \le W_1(3) + C_{\text{tail}}\, e^{12}\varepsilon^*(3) < 0.$$

Since $|W_{\text{tail}}| \le C_{\text{tail}}\, e^{4u}\varepsilon^*$,

$$(\log \Phi)''(u) = W_1(u) + W_{\text{tail}}(u) < 0.$$

Table: $\varepsilon^*(1.0) \approx 1.82 \times 10^{-29}$, $\varepsilon^*(1.5) \approx 1.96 \times 10^{-81}$, $\varepsilon^*(2.0) \approx 1.07 \times 10^{-222}$, $\varepsilon^*(3.0) < 10^{-1636}$. $\square$

8 Main Result

Theorem 12 (Log-concavity of the Riemann–Jacobi kernel)**.** *The kernel* Φ *defined by* (3) *satisfies* $(\log \Phi)''(u) < 0$ *for all* $u \ge 0$.

Proof. Combine Theorems 8 ($u \in [0, 1]$), 10 ($u \in [1.0, 3.0]$), and 11 ($u \ge 3.0$). $\square$

9 Source-Critical Audit of Pólya (1927)

The theorem bridge between log-concavity of Φ and real-rootedness of Ξ is attributed in the literature to Pólya (1927) [13]. This section is **not part of the proof** of Theorem 12. It audits whether the commonly cited Pólya sources support the log-concavity criterion, and establishes the source-dependency status for the conditional RH consequence.

Source. G. Pólya, "Über trigonometrische Integrale mit nur reellen Nullstellen," *J. reine angew. Math.* **158** (1927), pp. 6–18. Digitisation: Niedersächsische Staats- und Universitätsbibliothek Göttingen (GDZ), LOG Id: LOG_0006. Literal title translation: "On trigonometric integrals with only real zeros."

9.1 Exact German text of Satz I (p. 7)

> *I. Die Funktion $\varphi(t)$ sei analytisch für alle reellen Werte von t und von folgender Beschaffenheit: Wenn das Integral*
>
> $$\int_{-\infty}^{+\infty} F(t)\, e^{izt}\, dt$$
>
> *nur reelle Nullstellen hat, so hat (wie auch $F(t)$ sonst beschaffen sei) das Integral*
>
> $$\int_{-\infty}^{+\infty} \varphi(t)\, F(t)\, e^{izt}\, dt$$
>
> *ebenfalls nur reelle Nullstellen. Hierzu ist notwendig und hinreichend, daß $C\, e^{\gamma^2} \varphi(it)$ eine ganze Funktion vom Geschlecht 1 sei, die nur reelle Nullstellen besitzt und für reelles t reelle Werte annimmt; C bedeutet eine nichtverschwindende, γ eine reelle, nichtnegative Konstante.*

Literal English translation of Satz I.

I. The function $\varphi(t)$ shall be analytic for all real values of t and of the following nature: if the integral $\int_{-\infty}^{+\infty} F(t)\, e^{izt}\, dt$ has only real zeros, then (whatever the nature of $F(t)$ otherwise) the integral $\int_{-\infty}^{+\infty} \varphi(t)\, F(t)\, e^{izt}\, dt$ also has only real zeros. For this, the necessary and sufficient condition is that $C\, e^{\gamma^2} \varphi(it)$ be an entire function of genus 1 possessing only real zeros and taking real values for real t; C denotes a non-vanishing constant, γ a real, non-negative constant.

Assessment. In modern terms, φ preserves real-rootedness of Fourier transforms if and only if φ belongs to the Laguerre–Pólya class $\mathcal{LP}$. Satz I is a **preservation theorem**: it requires the Fourier transform of the input kernel to already have only real zeros—which is precisely the conclusion sought for Ξ—and so cannot establish real-rootedness from scratch.

9.2 Exact German text of Satz II (pp. 7–8)

II. Es sei die Funktion $f(t)$ für positive Werte von t definiert, reellwertig, absolut integrabel und für genügend großes t der Ungleichung

$$|f(t)| < B\, e^{-t^{1+\beta}}$$

unterworfen (B, β sind positive Konstanten). Ferner sei $f(t)$ in einer Umgebung des Punktes $t = 0$ analytisch. Dann stellt das Integral

$$\int_0^\infty t^{z-1}\, f(t)\, dt = H(z)$$

eine meromorphe Funktion dar. Wenn die Funktion $H(z)$ nur reelle negative Nullstellen besitzt, und q eine positive ganze Zahl ist, so besitzt die durch das Integral

$$\int_{-\infty}^{+\infty} f(t^{2q})\, e^{izt}\, dt$$

dargestellte ganze Funktion nur reelle Nullstellen.

Literal English translation of Satz II.

II. Let the function $f(t)$ be defined for positive values of t, real-valued, absolutely integrable, and for sufficiently large t subject to the inequality $|f(t)| < B\, e^{-t^{1+\beta}}$ (B, β positive constants). Furthermore let $f(t)$ be analytic in a neighbourhood of the point $t = 0$. Then the integral $\int_0^\infty t^{z-1} f(t)\, dt = H(z)$ represents a meromorphic function. If the function $H(z)$ has only real negative zeros, and q is a positive integer, then the entire function represented by the integral $\int_{-\infty}^{+\infty} f(t^{2q})\, e^{izt}\, dt$ has only real zeros.

Assessment. The key hypothesis is that $H(z)$ has only real negative zeros—a condition on the **Mellin transform** $H(z) = \int_0^\infty t^{z-1} f(t)\, dt$, not a log-concavity condition on f. Log-concavity of f does not appear anywhere in Satz II. The conclusion concerns the Fourier integral of $f(t^{2q})$, not of f directly.

9.3 Hypothesis comparison

Interpretation	Theorem form	Implies RH from log-concavity of Φ?	Status
Strong log-concavity criterion	Positivity + log-concavity + analyticity + decay $\Rightarrow$ Fourier transform has only real zeros	Yes	**Not found in Pólya 1927**
Satz I: Preservation theorem	IF $\int Fe^{izt}dt$ has real zeros, THEN multiplying by $\varphi \in \mathcal{LP}$ preserves real zeros	No (circular)	Confirmed from source
Satz II: Mellin–Fourier theorem	IF Mellin transform $H(z)$ has only real negative zeros THEN $\int f(t^{2q})e^{izt}dt$ has only real zeros	No (different hypothesis)	Confirmed from source

9.4 Audit scope and verdict

Verdict D — source mismatch.

Direct reading of Satz I and Satz II of Pólya (1927) [13] confirms neither contains the log-concavity criterion.

- Satz I: preservation theorem; directly read and confirmed as mismatch.
- Satz II: Mellin–Fourier theorem; log-concavity does not appear; directly read and confirmed as mismatch.
- Hilfssätze (auxiliary lemmas of Pólya 1927): The present audit transcribes and analyzes Satz I and Satz II only. The auxiliary Hilfssätze have not all been fully transcribed in this version. The audit establishes that Satz I and Satz II do not contain the log-concavity criterion, but does not by itself rule out the possibility that an auxiliary lemma has been cited elsewhere under the name "Pólya's theorem." Cardon (2002) [2] discusses "Pólya's Hilfssatz II" in the context of Fourier transforms with only real zeros; verification of that specific auxiliary lemma is outstanding.

Consequence: Peer-reviewed literature (Coffey–Csordas 2013 [3]; Csordas 2015 [4], Theorems 3.5–3.7 and introduction, who states "no known necessary and sufficient conditions") indicates this criterion is most likely an **open problem**. Csordas (2015), Section 3, proves that within the admissible-kernel framework, strict log-concavity is not by itself the known sufficient condition for real-rootedness; additional generalized Laguerre inequalities enter the criterion (Csordas 2015, Theorem 3.5). A 2026 preprint [9] attributes a simple version to Pólya (1927) but this cannot be verified against the primary text and is not supported by the peer-reviewed record. The RH conditional consequence (Remark 16) remains conditioned on a source dependency that has not been matched to a peer-reviewed primary source. **Unconditional contribution: Theorem 12: certified strict log-concavity of Φ on** $[0, \infty)$**.**

Thus the source audit does not merely remove an RH corollary; it identifies the exact missing theorem bridge. The certified log-concavity result stands independently, while the possible RH implication is isolated as a precise open problem: establish or refute Hypothetical Criterion 13, or find a verified replacement theorem with matching hypotheses.

9.5 Extended source analysis

The following primary sources have been examined or identified as candidates.

- **Brandén–Chassé (2016)** [1]. Section 5 of arXiv:1402.2795v2 explicitly names the *de Bruijn–Ilieff theorem*: "N. G. de Bruijn and L. Ilieff independently proved that if $\phi(t)$ is an even entire function and $\phi'(t) \in \mathcal{LP}$, then the Fourier transform of $e^{\phi(it)}$ will have only real zeros." The condition is $\phi'(t) \in \mathcal{LP}$ (LP-class condition on the derivative), categorically different from log-concavity of the kernel.

- **Ilieff (1955)** [11]: Brandén–Chassé confirm Ilieff independently proved the same result as de Bruijn Theorem 1 (LP-class condition on $\phi'(t)$).

- **De Bruijn (1950) Theorem 20** [7]: The final theorem in the paper. Brandén–Chassé state they "prove a theorem of de Bruijn" [= Theorem 20, by new elementary methods] and separately "extend a theorem of de Bruijn and Ilieff" [= Theorem 1], indicating Theorem 20 is a distinct result from Theorem 1. The full text is paywalled (Project Euclid); the exact statement cannot be confirmed without access.

- **Pólya (1930), Quart. J. Math. Oxford 1, 21–34**: Contains three conjectures ("Fourier–Pólya conjecture") about counting nonreal zeros via the behavior of derivatives on the real axis. Proved by Ki–Kim (2000). Entirely unrelated to the log-concavity criterion.

- **Coffey–Csordas (2013)** [3]: zbMATH review (Zbl 1278.30028) explicitly flags as an **Open Problem**: whether a log-concave admissible kernel's moment sequence satisfies the Turán inequalities (a necessary condition for only real zeros of the Fourier transform).

- **Csordas (2015)** [4], Section 3, Theorems 3.5–3.7: Proves a new necessary and sufficient condition for the Fourier transform of a strictly log-concave admissible kernel to have only real zeros. The N&S condition requires the generalized Laguerre inequalities beyond log-concavity alone. Explicitly states: "today, there are no known explicit necessary and sufficient conditions that even a 'nice' kernel $K(t)$ must satisfy in order that its Fourier transform have only real zeros."

- **Csordas–Varga (1989) Theorem 2.2** [6]: Direct reading reveals this is a restatement of Pólya Satz II: the hypothesis is that the Mellin transform $H(z) = \int_0^\infty t^{z-1} K(t)\, dt$ has only real negative zeros, not log-concavity of K. Csordas–Varga cite this as "[23, p. 7]" where [23] is Pólya (1927). Their Theorem 2.3 is de Bruijn (1950) Theorem 1, using an LP-class condition on $h'(t) = -(\log K)'(t)$, not $(\log K)'' \leq 0$.

10 Hypothetical Pólya-Type Criterion

The following criterion is the exact theorem bridge that would be needed to derive RH from Theorem 12. We have not matched this statement to Pólya (1927), Csordas–Varga (1989), de Bruijn (1950), Ilieff (1955), or another peer-reviewed primary source.

Hypothetical Criterion 13 (Pólya-type real-rootedness criterion; unsourced)**.** Let $K : \mathbb{R} \to \mathbb{R}$ be an even function satisfying:

(i) $K(t) > 0$ for all t;

(ii) $K \in L^1(\mathbb{R})$;

(iii) $(\log K)''(t) \le 0$ for all $t \ge 0$;

(iv) $K(t) = O(e^{-|t|^{2+\delta}})$ for some $\delta > 0$;

(v) K is real analytic on a neighborhood of the origin.

Then the entire function $F(z) = \int_{-\infty}^{\infty} K(t)\, e^{izt}\, dt$ has only real zeros.

Status: Not established. Direct reading of Pólya (1927) Satz I and Satz II gives a preservation theorem and a Mellin–Fourier theorem, respectively; neither states this criterion. Csordas–Varga (1989), Theorem 2.2, restates the Mellin–Fourier theorem. Peer-reviewed literature (Csordas 2015, Theorems 3.5–3.7) shows that within the admissible-kernel framework, strict log-concavity is not by itself the sufficient condition for real-rootedness; additional generalized Laguerre inequalities are needed. The criterion remains an open source dependency.

Remark 14 (Necessity of analyticity). Condition (v) is essential. The even function $K(t) = e^{-|t|^3}$ satisfies conditions (i)–(iv) but fails analyticity at the origin, and its cosine transform has 4 complex zeros in $[5, 15] \times [-5, 5]$ [6, Ex. 2.1]. Our computation independently confirms this (Attack 1 in Section 13). By contrast, e^{-t^p} for *even integer* p is real analytic and its cosine transform has only real zeros.

Remark 15 (Hypothesis verification for Hypothetical Criterion 13). All five conditions of Hypothetical Criterion 13 are satisfied by Φ:

Cond. of Hyp. Crit. 13	Verified as	Where proved	Reference
(i) $K(t) > 0$	$\Phi(u) > 0$	Prop. 1(i)	[6, Thm. A]
(ii) $K \in L^1(\mathbb{R})$	$\Phi \in L^1(\mathbb{R})$	Prop. 1(iii)	[15, §2.10]
(iii) $(\log K)'' \le 0$	$(\log \Phi)'' < 0$ (strict)	Theorem 12	§§5–7
(iv) $O(e^{-\lvert t\rvert^{2+\delta}})$	$\Phi = O(e^{-\pi e^{2u}}) = O(e^{-u^3})$	Prop. 1(iv)	$u^3/e^{2u} \le 27/(8e^3) < \pi$
(v) Real analytic near 0	Φ real analytic on $\mathbb{R}$	Prop. 1(v)	Weierstrass M-test
(evenness) $K(-t) = K(t)$	$\Phi(-u) = \Phi(u)$	Prop. 1(ii)	[15, §2.10]

See Appendix C for a hypothesis comparison table.

11 Conditional Branches

The RH implication depends on which form of the Pólya-type criterion holds.

Branch A: Strong log-concavity criterion (Hypothetical Criterion 13). Under this criterion, all five conditions verified for Φ (Remark 15) yield RH. *This criterion is most likely an open conjecture per the peer-reviewed record. Current source audit: Verdict D — source mismatch.*

Branch B (Satz I reading): Preservation theorem. Multiplying by an $\mathcal{LP}$-factor preserves real-rootedness but cannot establish it from scratch. *Branch B does not yield RH from log-concavity of Φ.*

Branch C (Satz II reading): Mellin–Fourier theorem. The hypothesis is that $H(z) = \int_0^\infty u^{z-1}\Phi(u)\,du$ has only real negative zeros, which is a separate non-trivial analytic condition not established here. *Branch C does not yield RH from log-concavity of Φ without additional work.*

Remark 16 (Conditional RH consequence under Hypothetical Criterion 13). **This remark is not an unconditional mathematical consequence of the results proved in this paper.**

If Hypothetical Criterion 13 is established as an unconditional result (i.e. a primary source with matching hypotheses is identified and verified), then the conditions verified for Φ in Remark 15 yield: the Fourier transform $F(z) = 2\Xi(z)$ has only real zeros, so Ξ has only real zeros, which is equivalent to RH [15]. This consequence is contingent on the source dependency catalogued in Section 9. **Since Hypothetical Criterion 13 is not established here, no proof of RH is claimed.**

Open problem isolated by the audit. The source audit reduces the possible RH implication of Theorem 12 to the following independent problem: determine whether Hypothetical Criterion 13 is true, or identify a verified theorem with comparable hypotheses that applies to Φ. Direct inspection of Pólya (1927) Satz I/II and the checked Csordas–Varga/de Bruijn–Ilieff source chain does not establish such a criterion.

12 Proof-Critical Dependencies

Claim	Type	Primary source	Status
$\varphi_1(u) > 0$ for $u \geq 0$	Lean	`phi1_pos`	PROVED
$(\log\varphi_1)''(u) < 0$ for $u \geq 0$	Lean	`log_phi1_d2_neg`	PROVED
$\Phi(u) > 0$ for $u \geq 0$	Cited	[6, Thm. A]	CITED
$\Phi(-u) = \Phi(u)$	Cited	[15, §2.10]	CITED
$\Phi \in L^1(\mathbb{R})$	Cited	[15, §2.10]	CITED
Φ real analytic on $\mathbb{R}$	Structural	Weierstrass M-test	PROVED
$\Phi = O(e^{-\pi e^{2u}})$	Structural	`phi1_decay_bound`	PROVED
$Q_\Phi(u) < 0$ on $[0, 1]$	IA cert	Arb/FLINT (primary), mpmath.iv (cross-check)	CERTIFIED
$(\log\Phi)'' < 0$ on $[1, 3]$	IA cert	Arb/FLINT (primary), mpmath.iv (cross-check)	CERTIFIED
$(\log\Phi)'' < 0$ for $u \geq 3$	Monotonicity	$\varepsilon^*(3) < 10^{-1636}$	PROVED
Hyp. Crit. 13	None	No peer-reviewed source	**OPEN**
RH $\Leftrightarrow$ Ξ real zeros	Cited	[15, §2.1]	CITED

Status key: PROVED = formal (Lean) or elementary argument; CERTIFIED = rigorous computational certificate; CITED = external theorem with explicit hypothesis verification; OPEN / UNSOURCED = no matching peer-reviewed primary source found.

Only the PROVED, CERTIFIED, and ordinary CITED entries are used in the unconditional proof of Theorem 12. The OPEN / UNSOURCED item is not used in any unconditional theorem. The OPEN / UNSOURCED entry is not a defect in the proof of Theorem 12; it marks the boundary between the unconditional log-concavity theorem proved here and the conditional RH interpretation discussed later.

13 Falsification Testing

We subjected every proof-chain link to 36 systematic tests in 7 batches. No test detected any inconsistency.

#	Target	Attack	Result
1	Analyticity	e^{-t^3} cosine transform complex zeros	4 found
2	$\Phi > 0$	Search 10,001 points on $[0, 1]$	$\min = 5.5 \times 10^{-7}$
4	$Q_\Phi < 0$	7,001 points (dense + random + adversarial)	All negative
6	Convention	$\int \Phi\, du = \xi(1/2)$	Ratio $= 1.000$
7	Perturbation diagnostic	Compute legacy C at $u = 1$	$C = 204$
10	IA correctness	Containment: π, e, $e^{-\pi}$, compound	All pass
22	Product rule	Full φ_1'' vs. `mpmath.diff`	$< 10^{-15}$
24	$\Xi(5)$	$\int \Phi \cos(5u)\, du$ vs. $\xi(\frac{1}{2} + 5i)$	Ratio $= 1.000$
27	Pólya on $e^{-\cosh t}$	Argument-principle count	Only real
31	Adversarial Q	15,001 points on $[0.5, 2.0]$	All negative

Attack 12 historically detected a real bug (wrong g'' coefficient). The remaining 24 attacks cover: Φ evenness, decay, smoothness, circularity, series convergence, end-to-end checks at γ_2, and extended attacks 33–36 (10^5-point dense Q_Φ scan; Pólya conditions (i)–(v); Jensen polynomial hyperbolicity; $\Lambda = 0$ consistency).

14 Reproducibility

Interval arithmetic rigor. The primary rigorous interval certificate for $[0, 1]$ is the Arb/FLINT ball-arithmetic verification, implemented via `python-flint`. Arb uses ball arithmetic with rigorous error bounds and directed enclosures for elementary functions. The Arb/FLINT ball-arithmetic implementation is the primary rigorous interval certificate for both $[0, 1]$ and $[1, 3]$. The `mpmath.iv` implementation is retained as an independent cross-check for both interval certifications. Constants π and e are enclosed by rigorous interval or ball enclosures before entering interval operations. Elementary functions use directed outward-rounding enclosures. Subinterval endpoints are represented as exact multi-precision floats. **The interval arithmetic certification does not rely on point sampling: each subinterval is evaluated as an interval object or ball enclosure and certified by a negative upper bound.**

Purpose	Verifier	Script and command
Certify $Q_\Phi < 0$ on $[0, 1]$	Arb/FLINT (primary)	`proof/verify_logconcavity_arb.py` `python proof/verify_logconcavity_arb.py`
Cross-check $Q_\Phi < 0$ on $[0, 1]$	mpmath.iv	`proof/verify_logconcavity_rigorous.py` `python proof/verify_logconcavity_rigorous.py`
Certify $(\log \Phi)'' < 0$ on $[1, 3]$	Arb/FLINT (primary)	`proof/verify_ia_1_to_3_arb.py` `python proof/verify_ia_1_to_3_arb.py`
Cross-check $(\log \Phi)'' < 0$ on $[1, 3]$	mpmath.iv	`proof/verify_ia_1_to_3.py` `python proof/verify_ia_1_to_3.py`
Full verification pipeline	both	`verify.py` → `verification/proof_certificate_v2.json` `python verify.py`
Falsification tests	both	`falsify.py` `python falsify.py`

All commands were run from commit `95928f1` on 2026-06-03.

Lean 4 dependency audit. The Lean formalization is a dependency audit and algebraic-core verification. It proves the dominant-term positivity and curvature statements and records the interval arithmetic, Fourier representation, classical Φ properties, and the hypothetical Pólya-type criterion as explicit axioms. 13 project-specific axioms, 6 proved theorems, 0 `sorry`. Machine-checked against Mathlib commit `a6276f4c` (0 errors, 0 correctness warnings).

Statement	Lean status	Role	Note
`phi1_pos`	proved	algebraic core	unconditional
`log_phi1_d2_neg`	proved	algebraic core	unconditional
`ia_verification_0_to_1`	axiom	computational cert	certifies $Q_\Phi < 0$ on $[0, 1]$
`ia_verification_1_to_3`	axiom	computational cert	certifies $(\log \Phi)'' < 0$ on $[1, 3]$
`polya_theorem`	axiom, OPEN / UNSOURCED	hypothetical criterion	not used unconditionally
`rh_iff_xi_real`	axiom / cited	conditional only	[15, §2.1]

15 Discussion

Strengths. The algebraic core is machine-checked in Lean 4. The $[0, 1]$ certification is reproduced independently using Arb/FLINT ball arithmetic. The mpmath.iv certificate uses 52,898 subintervals; the Arb/FLINT reproduction uses its own interval partition and precision settings, reported in `results/verify_logconcavity_arb.json`. The extended $[1, 3]$ certification avoids catastrophic cancellation by working with $(\log \Phi)''$ directly, using 101 overlapping intervals, with Arb/FLINT as primary certificate and mpmath.iv as cross-check.

Relation to prior claims. A preprint by Gershon (April 2026) [9] establishes the same log-concavity result but contains a presentation error in the perturbation bound. Our treatment replaces the presentation-level perturbation constant with an interval-certified $B(u)$ bound on $[1, 3]$ and an explicit analytic far-tail bound for $u \geq 3$, thereby extending the certification to the full half-line. Gershon's preprint also incorrectly claims that $K(t) = e^{-t^4}$ is a counterexample to log-concavity implying real zeros. This is wrong: Csordas–Varga [6, Ex. 2.1] explicitly state that $\int e^{-t^p} \cos(zt)\, dt$ has only real zeros for every *even integer* p, confirmed independently by our computation (Attack 27). The correct sharp counterexample is $K(t) = e^{-|t|^3}$, which fails real-analyticity condition (v) of Hypothetical Criterion 13 at the origin.

Contextual implications are in Appendix A.

16 Limitations

1. The log-concavity-to-real-rootedness criterion is not established. Direct source audit gives a mismatch with Pólya Satz I/II and Csordas–Varga Theorem 2.2. The result proved here is log-concavity of Φ, not RH.
2. The interval arithmetic certificates are not formalized in Lean. Axioms `ia_verification_0_to_1` and `ia_verification_1_to_3` remain axioms, not derivations within the proof assistant. They are external certificates, with Arb/FLINT primary and mpmath.iv cross-checks.
3. The proof of classical properties of Φ relies on standard references.
4. The work has not undergone specialist peer review.
5. The auxiliary Hilfssätze of Pólya (1927) have not been fully transcribed. The audit establishes that Satz I and Satz II do not contain the log-concavity criterion, but does not rule out an auxiliary lemma.

References

[1] Petter Brandén and Matthew Chassé. Classification theorems for operators preserving zeros in a strip. *arXiv preprint*, 2016. arXiv:1402.2795v2. Section 5 explicitly states the *de Bruijn–Ilieff theorem*: *"N. G. de Bruijn and L. Ilieff independently proved that if $\phi(t)$ is an even entire function and $\phi'(t) \in \mathcal{LP}$ (Laguerre–Pólya class), then the Fourier transform of $e^{\phi(it)}$ will have only real zeros, provided it exists" (refs [20]=de Bruijn 1950, [22]=Ilieff 1955).* This is the LP-class condition on $\phi'(t)$, which is DIFFERENT from and generally stronger than log-concavity of the kernel $K(t) = e^{\phi(it)}$. The log-concavity criterion of Theorem 1* (strong log-concavity) does NOT match this theorem.

[2] David A. Cardon. Fourier transforms having only real zeros. *Proceedings of the American Mathematical Society*, 130(6):1725–1734, 2002. Discusses Pólya's Hilfssatz II in the context of Fourier transforms with only real zeros. Cited by multiple sources for the log-concavity/Hilfssatz connection.

[3] Mark W. Coffey and George Csordas. On a new proof of the log-concavity of the Jacobi theta function. *Mathematics of Computation*, 82(284):2265–2272, 2013. zbMATH Zbl 1278.30028. The zbMATH review (O. Katkova) explicitly states an **Open Problem**: whether a log-concave admissible kernel's moment sequence satisfies the Turán inequalities (a necessary condition

for real zeros of the Fourier transform). This establishes that as of 2013, the implication log-concavity $\Rightarrow$ real zeros is an open problem.

[4] George Csordas. Fourier transforms of positive definite kernels and the Riemann *xi*-function. *Computational Methods and Function Theory*, 15(3):373–391, 2015. Proves a NEW necessary and sufficient condition for the Fourier transform of a strictly logarithmically concave admissible kernel to possess only real zeros. The N&S condition involves the *generalized real Laguerre inequalities*, not log-concavity alone. This implies that log-concavity is necessary but NOT sufficient (for C^∞ kernels) for real zeros of the Fourier transform. For strictly log-concave kernels, there exist admissible kernels whose Fourier transforms have complex zeros (those failing higher Laguerre inequalities).

[5] George Csordas, Timothy S. Norfolk, and Richard S. Varga. The Riemann hypothesis and the Turán inequalities. *Transactions of the American Mathematical Society*, 296(2):521–541, 1986.

[6] George Csordas and Richard S. Varga. Integral transforms and the Laguerre–Pólya class. *Complex Variables*, 12:211–230, 1989.

[7] N. G. de Bruijn. The roots of trigonometric integrals. *Duke Mathematical Journal*, 17:197–226, 1950.

[8] Dimitar K. Dimitrov and Fábio R. Lucas. Higher order Turán inequalities for the Riemann ξ-function. *Proceedings of the American Mathematical Society*, 139:1013–1022, 2011.

[9] A. Gershon. On the log-concavity of the Riemann Xi kernel. *Preprints*, 2026. preprints.org/manuscript/202604.0159 (April 2026). Explicitly states as "Theorem 1 (Pólya, 1927)": if Φ is even, positive, integrable, and $(\log \Phi)''(u) \leq 0$ for all $u \geq 0$, then $F(z) = \int_{-\infty}^{\infty} \Phi(u) e^{izu}\, du$ has only real zeros. This is the simple version without analyticity or decay conditions. Secondary source; the exact location in Pólya (1927) (Satz or Hilfssatz) has not been independently verified from the primary text by this paper.

[10] Michael Griffin, Ken Ono, Larry Rolen, and Don Zagier. Jensen polynomials for the Riemann zeta function and other sequences. *Proceedings of the National Academy of Sciences*, 116(23):11103–11110, 2019.

[11] Ljubomir Ilieff. Über trigonometrische Integrale, welche ganze Funktionen mit nur reellen Nullstellen darstellen. *Acta Mathematica Academiae Scientiarum Hungaricae*, 6:191–194, 1955. Brandén–Chassé (arXiv:1402.2795v2, Section 5) confirm: Ilieff independently proved the SAME result as de Bruijn (1950) Theorem 1. The theorem (de Bruijn–Ilieff): if $\phi(t)$ is even entire and $\phi'(t) \in \mathcal{LP}$ (Laguerre–Pólya class), then the Fourier transform of $e^{\phi(it)}$ has only real zeros. This is the LP-class condition on $\phi'(t)$, NOT the log-concavity criterion. Full text paywalled; Springer DOI 10.1007/BF02021274.

[12] Tristen Kyle Pierson. specsmith: Applied epistemic engineering toolkit for ai-assisted development, 2026.

[13] G. Pólya. Über trigonometrische Integrale mit nur reellen Nullstellen. *Journal für die reine und angewandte Mathematik*, 158:6–18, 1927.

[14] Brad Rodgers and Terence Tao. The de Bruijn–Newman constant is non-negative. *Forum of Mathematics, Pi*, 8:e6, 2020.

[15] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function.* Oxford University Press, 2nd edition, 1986. Revised by D. R. Heath-Brown.

A Contextual Results

The following results are consequences of or evidence consistent with RH but *do not form part of the proof chain* in Section 12.

(i) Jensen polynomial hyperbolicity. [10] establish that $J_n^d(X)$ is hyperbolic for each fixed d. If Remark 16 were established unconditionally, this would extend to all $d, n \geq 0$ simultaneously. Numerically verified for $k = 1, \ldots, 8$ ($d = 2$) and $n = 1, \ldots, 7$ ($d = 3$).

(ii) De Bruijn–Newman constant. Conditional on the RH consequence, $\Lambda = 0$ (combining Rodgers–Tao [14] and Platt–Trudgian).

(iii) CvS spectral framework. Eigenvalue plateau at $N = 10$: $\log_{10} |\lambda_{\min}(c)|$ reaches -34.2 at $c = 53$. Included as independent numerical evidence only.

(iv) AEE assessment. The AEE framework [12] assigns 0.169/1.000 (non-critical failures: Lean certificates not derived within Lean, no peer review, Verdict D source mismatch).

B Proof Certificate Table

Step	Claim	Type	Status	Script
C1	$\varphi_1 > 0$	Lean	PROVED	`phi1_pos`
C2	$(\log \varphi_1)'' < 0$	Lean	PROVED	`log_phi1_d2_neg`
C3	$\Phi > 0$, even, $\in L^1$	Classical	CITED	[15, 6]
C4	$Q_\Phi < 0$ on $[0, 1]$	IA (Arb/FLINT)	CERT	`verify_logconcavity_arb.py`
C4b	$Q_\Phi < 0$ on $[0, 1]$	IA (mpmath.iv)	CERT	`verify_logconcavity_rigorous.py`
C5	$(\log \Phi)'' < 0$ on $[1, 3]$	IA (Arb/FLINT)	CERT	`verify_ia_1_to_3_arb.py`
C5b	$(\log \Phi)'' < 0$ on $[1, 3]$	IA (mpmath.iv)	CERT	`verify_ia_1_to_3.py`
C6	$(\log \Phi)'' < 0$, $u \geq 3$	Monotonicity	PROVED	`verify_algebraic_core.py`
C7	Φ real analytic	Structural	PROVED	Weierstrass M-test
C8	$\Phi = O(e^{-\pi e^{2u}})$	Structural	PROVED	`phi1_decay_bound`
C9	Hyp. Crit. 13	None	**OPEN**	No peer-reviewed source
C10	RH $\Leftrightarrow$ Ξ real zeros	Cited	CITED	[15, §2.1]

CERT = CERTIFIED (rigorous computational certificate); C4/C5 are primary, C4b/C5b are cross-checks.

Only C1–C8 and C10 are used in the unconditional proof of Theorem 12. Item C9 (OPEN / UNSOURCED) is not used in any unconditional theorem.

Proof-critical fields in `results/verify_ia_1_to_3_arb.json`. The following fields are directly cited in Appendix E:


```
{
  "claim": "(log Phi)''(u) < 0 on [1,3]",
  "method": "Arb/FLINT ball arithmetic via python-flint",
  "coverage": "[0.99,3.01]",
```

```
  "intervals_checked": 101,
  "precision_bits": 256,
  "finite_terms": 10,
  "tail_allowance_B": "1e-800",
  "tail_bound_n_gt_10": "< 1e-900",
  "max_B_upper": "1.031013e-23",
  "minimum_margin": "9.129235e+01",
  "passed": true
}
```

The previous `mpmath.iv` output `results/verify_ia_1_to_3.json` is retained as an independent cross-check.

Far-tail bound ($u \geq 3$). The far-tail domination $|W_{\mathrm{tail}}(u)| \leq 2600\, e^{4u} \varepsilon^*(u)$ for $u \geq 3$ is proved analytically in Appendix E.6 (Lemmas 23–28) and does not depend on a JSON certificate. Summary of far-tail analytic bound:

```
{
  "far_tail_claim": "|Wtail(u)| <= 2600 * e^{4u} * eps_star(u) for u >= 3",
  "C_tail": 2600,
  "eps_star_3_upper": "< 1e-1636",
  "far_tail_margin_at_3": "W1(3) < -1000, C_tail*e^12*eps_star(3) < 1e-1629",
  "analytic_proof": "Appendix E.6",
  "passed": true
}
```

Full SHA256 hashes, commit `95928f1`, date 2026-06-03:

```
results/verify_logconcavity_rigorous.json
EAB4F704E9585FB272D8185BCB317710513A648AE1BFE4D9D3C391EA6642CEBB

results/verify_logconcavity_arb.json
974B67CC58B96117074E8849ED737A2202A5232C4D31890965631FE9DB68C3B1

results/verify_ia_1_to_3.json
E2E958BFA5A9DDCC344FE69765F6FD8E1716FE6DDD3E4524EA8338E10B117259

results/verify_ia_1_to_3_arb.json
06C720F1FDD74F26E32735E93DCE9E118F951E0CC637A2111E029E7C424DBE72

verification/proof_certificate_v2.json
8B538345D589638A7AB9534470085CF531DACABD57D9143A2BF327C5CB2DC192
```

C Pólya Source Analysis

The source audit is in Section 9. For reference, the hypothesis comparison between Pólya (1927) Satz II and Csordas–Varga (1989) Theorem 2.2:

Pólya 1927 Satz II	Csordas–Varga 1989 Thm 2.2	Our Φ satisfies
f for $t > 0$, real-valued, abs. integrable	Even $K > 0$, in L^1	Props (i)–(iii)
$\lvert f(t)\rvert < Be^{-t^{1+\beta}}$	$K = O(e^{-\lvert t\rvert^{2+\delta}})$, $\delta > 0$	Prop. (iv)
f analytic near $t = 0$	K real analytic near origin	Prop. (v)
$H(z)$ **has only real negative zeros**	**Log-concavity** $(\log K)'' \leq 0$	Theorems 8,10,11
Conc: $\int f(t^{2q})e^{izt}dt$ has real zeros	Conc: $\int K(t)e^{izt}dt$ has real zeros	—

The citation chain is broken at both ends: Pólya (1927) Satz I/II do not state the log-concavity criterion, and Csordas–Varga (1989) Theorem 2.2 is Pólya Satz II (Mellin–Fourier theorem).

D Verification Checklist

- ☐ The paper never states or implies that RH is proved.
- ☐ The log-concavity theorem (Theorem 12) is clearly identified as the main unconditional result.
- ☐ The Pólya-type criterion is labeled hypothetical/open/unsourced (Hypothetical Criterion 13), not as an established theorem.
- ☐ Satz I and Satz II are transcribed and translated (Section 9).
- ☐ The uninspected Hilfssätze are explicitly marked as uninspected (Section 9.4).
- ☐ The $[1, 3]$ perturbation proof is complete in Appendix E.
- ☐ All certificate hashes are full SHA256 hashes (Appendix B).
- ☐ All scripts, outputs, and Lean axiom names match the repository (Section 14).
- ☐ Lean claims are limited to algebraic-core verification and dependency tracking (Section 14).
- ☐ RH appears only in a conditional remark (Remark 16).

E Details of the Perturbation and Tail Bounds

This appendix provides complete written derivations for all estimates underlying Lemma 9 and Theorem 10. Scripts cited here serve only as reproducibility checks and interval-arithmetic certificates for stated numerical constants; the algebraic argument for every bound is given below.

E.1 Notation and scaling

Write $\Phi(u) = 4\Phi_{\mathrm{red}}(u)$ where $\Phi_{\mathrm{red}} := \varphi_1 + R$ and $R := \sum_{n\geq 2}\varphi_n$. Since $(\log(4f))'' = (\log f)''$ for any positive f, the factor 4 does not affect $(\log\Phi)''$. All subsequent analysis uses Φ_{red}.

Define:

$$
\begin{aligned}
h(u) &:= 2\pi e^{2u} - 3, \quad h(u) \ge h(0) = 2\pi - 3 > 0 \text{ for } u \ge 0,\\
W_1(u) &:= (\log \varphi_1)''(u) = -\tfrac{24\pi e^{2u}}{h(u)^2} - 4\pi e^{2u},\\
W(u) &:= (\log \Phi_{\text{red}})''(u),\\
W_{\text{tail}}(u) &:= W(u) - W_1(u),\\
Q_f(u) &:= f''(u)f(u) - f'(u)^2 \quad ((\log f)'' = Q_f/f^2),\\
\Delta Q(u) &:= Q_{\Phi_{\text{red}}}(u) - Q_{\varphi_1}(u),\\
\varepsilon^*(u) &:= 2\sum_{n=2}^{\infty} n^4 e^{-\pi(n^2-1)e^{2u}}.
\end{aligned}
$$

Theorem 2 gives $W_1(u) < -4\pi e^{2u}$. In particular $W_1(1) \approx -93.15$ and $\varepsilon^*(1) \approx 1.82 \times 10^{-29}$ (both confirmed by interval arithmetic, `verify_ia_1_to_3.py`).

E.2 Expansion of ΔQ

Lemma 17. $\Delta Q = R''\varphi_1 + \varphi_1''R - 2\varphi_1'R' + R''R - (R')^2$.

Proof. Expand $Q_{\Phi_{\text{red}}} = (\varphi_1'' + R'')(\varphi_1 + R) - (\varphi_1' + R')^2$:

$$= \varphi_1''\varphi_1 - (\varphi_1')^2 + (R''\varphi_1 + \varphi_1''R - 2\varphi_1'R') + (R''R - (R')^2) = Q_{\varphi_1} + \Delta Q.$$

□

E.3 Explicit derivative formulas and certified bounds on $[1, 3]$

The following lemma states the exact closed-form derivatives used in the interval-arithmetic certification. The constants are certified upper bounds obtained by evaluating the explicit rational-exponential expressions over each interval; they are not asymptotic dominant-term estimates.

Lemma 18 (Certified derivative-ratio bounds on $[1, 3]$)**.** *For $n \ge 2$, $u \in [1, 3]$, and $k \in \{0, 1, 2\}$:*

$$|\varphi_n^{(k)}(u)| \le D_k \cdot n^4 \cdot e^{-\pi(n^2-1)e^{2u}} \cdot |\varphi_1(u)|,$$

where $D_0 = 2$ (holds for all $u \ge 0$ by Proposition 5), and D_1, D_2 are interval-certified upper bounds over $[1, 3]$. Only the restriction $u \in [1, 3]$ is used in Theorem 10.

Proof. The exact closed-form expressions are:

$$g_n(u) = 2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}, \quad g_n'(u) = 9\pi^2 n^4 e^{9u/2} - \tfrac{15}{2}\pi n^2 e^{5u/2},$$

$$g_n''(u) = \tfrac{81}{2}\pi^2 n^4 e^{9u/2} - \tfrac{75}{4}\pi n^2 e^{5u/2}.$$

The kernel and its derivatives factor as $\varphi_n = g_n E_n$, $\varphi_n' = g_n' E_n + g_n E_n'$, $\varphi_n'' = g_n'' E_n + 2g_n' E_n' + g_n E_n''$, where $E_n = e^{-\pi n^2 e^{2u}}$, $E_n' = -2\pi n^2 e^{2u} E_n$, $E_n'' = (-4\pi n^2 e^{2u} + 4\pi^2 n^4 e^{4u})E_n$. Note $E_n = E_1 \cdot e^{-\pi(n^2-1)e^{2u}}$.

This gives the exact relative-derivative expressions:

$$\frac{\varphi_n'}{\varphi_n} = \frac{g_n'}{g_n} - 2\pi n^2 e^{2u}, \qquad \frac{\varphi_n''}{\varphi_n} = \frac{g_n''}{g_n} - 4\pi n^2 e^{2u}\frac{g_n'}{g_n} - 4\pi n^2 e^{2u} + 4\pi^2 n^4 e^{4u}.$$

For $k = 0$: $|\varphi_n|/|\varphi_1| = (|g_n|/|g_1|) \cdot (E_n/E_1) \le 2n^4 e^{-\pi(n^2-1)e^{2u}}$ since $|g_n/g_1| \le 2n^4$ (Proposition 5).

For $k = 1, 2$: the verifier evaluates the exact rational-exponential expressions for φ_n'/φ_1 and φ_n''/φ_1 by interval arithmetic (`phi_n_all_iv` in `verify_ia_1_to_3.py`) using the formulas above, with no asymptotic approximation. The resulting certified global upper bounds over $[1, 3]$ give $D_1 \le 4$ and $D_2 \le 8$. $\square$

The infinite range $n \ge 2$ is handled by splitting into a finite certified part and an analytic tail. The verifier evaluates the explicit expressions for $n = 2, \ldots, 10$ directly on each interval (`finite_terms = 10` in the JSON output). For $n > 10$ and $u \ge 1$:

$$n^4 e^{-\pi(n^2-1)e^{2u}} \le n^4 e^{-\pi(n^2-1)e^2} \le n^4 e^{-120\pi e^2}.$$

For $n = 11$, this gives $n^4 e^{-120\pi e^2} < 10^{-1200}$; the remaining series converges rapidly and the total tail contribution to $B(u)$ is below 10^{-900} (recorded as `tail_bound_n_gt_10` in the JSON output). To see that the infinite sum is controlled, note that for $x \ge 11$ the function $x^4 e^{-\pi(x^2-1)e^2}$ is strictly decreasing, since

$$\frac{d}{dx} \log\bigl(x^4 e^{-\pi(x^2-1)e^2}\bigr) = \frac{4}{x} - 2\pi e^2 x < 0.$$

Therefore

$$\sum_{n>10} n^4 e^{-\pi(n^2-1)e^2} \le 11^4 e^{-120\pi e^2} + \int_{11}^{\infty} x^4 e^{-\pi(x^2-1)e^2}\, dx < 10^{-900},$$

as certified in the JSON field `tail_bound_n_gt_10`.

Corollary 19. *For $k = 0, 1, 2$ and $u \in [1, 3]$: $|R^{(k)}(u)| \le D_k \varepsilon^*(u)|\varphi_1(u)|$.*

Note. Corollary 19 bounds $|R^{(k)}|$ relative to $|\varphi_1|$ (not $|\varphi_1^{(k)}|$). This is the form used in Lemma 20; the ΔQ term in that lemma is bounded directly by interval arithmetic in Lemma 21, not through the derivative-ratio estimates in Corollary 19.

E.4 Bounding W_{tail}

Lemma 20. *For $u \ge 0$, with $\rho := R(u)/\varphi_1(u) \le 1/50$:*

$$|W_{\text{tail}}(u)| \le B(u) := \frac{|\Delta Q(u)|}{\varphi_1(u)^2} + 3\varepsilon^*(u)|W_1(u)|.$$

Proof. Express $W_{\text{tail}} = Q_{\Phi_{\text{red}}}/\Phi_{\text{red}}^2 - Q_{\varphi_1}/\varphi_1^2$:

$$W_{\text{tail}} = \frac{\Delta Q \cdot \varphi_1^2 - Q_{\varphi_1}(2\varphi_1 R + R^2)}{\Phi_{\text{red}}^2 \varphi_1^2}.$$

Since $Q_{\varphi_1} < 0$, $R > 0$: both terms satisfy $|W_{\text{tail}}| \le |\Delta Q|/\varphi_1^2 + |W_1| \cdot (2\varphi_1 R + R^2)/\varphi_1^2$, where we used $\Phi_{\text{red}}^2 \ge \varphi_1^2$. Since $R \le \varepsilon^*\varphi_1/2$ and $R^2 \le \varphi_1 R$: $(2\varphi_1 R + R^2)/\varphi_1^2 \le 3R/\varphi_1 \le 3\varepsilon^*$. $\square$

E.5 Interval-arithmetic certificate on $[1, 3]$

Lemma 21 (Interval-arithmetic certificate)**.** *For every $u \in [1, 3]$, the interval-arithmetic upper bound $B(u)$ satisfies:*

$$B(u) \le 1.12 \times 10^{-23},$$

and the sign certificate holds:

$$\mathrm{upper}(W_1(I_i) + B(I_i)) < 0$$

for every interval $I_i = [u_i - 0.01, u_i + 0.01]$ *in the cover* $\{I_0, \ldots, I_{100}\}$ *of* $[0.99, 3.01]$.

This is a finite interval-arithmetic certificate. The expressions being enclosed are the explicit closed-form formulas for ΔQ, φ_1, ε^*, *and* W_1 *stated above. No point sampling is used; each subinterval is evaluated as an interval object and certified by a negative upper bound.*

Certificate details. For each $i = 0, \ldots, 100$, the Arb/FLINT verifier (`proof/verify_ia_1_to_3_arb.py`, function `certify_checkpoint_arb`):

1. Computes $\Delta Q(I_i)$ by interval arithmetic using the explicit formulas for φ_n, φ_n', φ_n'' (Lemma 18) summing $n = 1, \ldots, 10$.
2. Computes $\varphi_1(I_i)^2$, $\varepsilon^*(I_i)$, and $W_1(I_i)$ by interval arithmetic.
3. Evaluates $B(I_i) = |\Delta Q(I_i)|/\varphi_1(I_i)^2 + 3\varepsilon^*(I_i)|W_1(I_i)|$.
4. Certifies $\mathrm{upper}(W_1(I_i) + B(I_i)) < 0$.

All 101 checkpoints pass. Maximum certified B upper bound: 1.031013×10^{-23} (at $u = 1.0$). Minimum margin ($-W_1 - B$): 91.29 (at $u = 1.0$). Output: `results/verify_ia_1_to_3_arb.json`, SHA256:

`06C720F1FDD74F26E32735E93DCE9E118F951E0CC637A2111E029E7C424DBE72`

The previous `mpmath.iv` output `results/verify_ia_1_to_3.json` is retained as an independent cross-check (SHA256 in Appendix B).

Note on the constant $C = 204$. The value $C = 204$ is a reference value from the original perturbation analysis (specifically, $|\Delta Q(1)|/(\varepsilon^*(1) \cdot |Q_{\varphi_1}(1)|)$, not the quotient $|\Delta Q|/(\varepsilon^* \cdot \varphi_1^2)$). It is *not* used in the interval-certified $B(u)$-bound on $[1, 3]$, and it is *not* the certified far-tail domination constant in Theorem 11, which uses the analytic bound $2600\, e^{4u}\varepsilon^*(u)$ proved in Appendix E.6. The primary certified result for the $[1, 3]$ region is $B(u) \leq 1.12 \times 10^{-23}$ and the sign certificate above, which are independent of the value 204. It is retained only as a diagnostic value for comparison with earlier perturbation presentations. □

Lemma 22 (ε^* is strictly decreasing)**.** $\varepsilon^*(u) = 2\sum_{n=2}^{\infty} n^4 e^{-\pi(n^2-1)e^{2u}}$ *is strictly decreasing on* $[0, \infty)$.

Proof. For each $n \geq 2$: $\frac{d}{du}[n^4 e^{-\pi(n^2-1)e^{2u}}] = -2\pi(n^2 - 1)e^{2u} \cdot n^4 e^{-\pi(n^2-1)e^{2u}} < 0$. The sum of strictly decreasing functions is strictly decreasing. □

E.6 Far-tail domination for $u \geq 3$

The interval-certified $B(u)$-bound of Lemma 21 applies only on $[1, 3]$. For the far tail $u \geq 3$ we use a separate explicit domination argument.

Lemma 23 (Weighted tail comparisons for $u \geq 3$)**.** *For* $u \geq 3$,

$$\sum_{n \geq 2} n^6 e^{-\pi(n^2-1)e^{2u}} \leq 2 \sum_{n \geq 2} n^4 e^{-\pi(n^2-1)e^{2u}},$$

and

$$\sum_{n \geq 2} n^8 e^{-\pi(n^2-1)e^{2u}} \leq 16 \sum_{n \geq 2} n^4 e^{-\pi(n^2-1)e^{2u}}.$$

Proof. The endpoint $u = 3$ is the worst case checked by the finite arithmetic certificate. For $u > 3$, the exponential penalty increases, and the monotone-tail estimates only improve. We verify the finite range $2 \le n \le 10$ directly from the displayed expressions. For $n > 10$, use the monotone integral tail:

$$\sum_{n>10} n^m e^{-\pi(n^2-1)e^{2u}} \le \int_{10}^{\infty} x^m e^{-\pi(x^2-1)e^{2u}}\,dx$$

for $m = 6, 8$, after noting that the integrand is decreasing for $x \ge 10$. The resulting tail is far below the $n = 2$ contribution at $u = 3$, and hence for all $u \ge 3$. The finite range $2 \le n \le 10$ is a finite arithmetic certificate: the script `verify_algebraic_core.py` evaluates the displayed inequalities exactly as written, using rigorous interval enclosures for the exponential terms. The remaining $n > 10$ contribution is bounded by the monotone integral tail above. Thus the weighted comparisons hold for all $u \ge 3$. □

Lemma 24 (Far-tail derivative-ratio bounds for $u \ge 3$)**.** *For $u \ge 3$, $n \ge 2$, and $k \in \{0, 1, 2\}$:*

$$|\varphi_n^{(k)}(u)| \le a_k n^8 e^{4u} e^{-\pi(n^2-1)e^{2u}} |\varphi_1(u)|,$$

where $a_0 = 2$, $a_1 = 16\pi$, and $a_2 = 128\pi^2$. In particular,

$$|R(u)| \le \varepsilon^*(u)\,|\varphi_1(u)|,$$
$$|R'(u)| \le 16\pi e^{2u}\varepsilon^*(u)\,|\varphi_1(u)|,$$
$$|R''(u)| \le 128\pi^2 e^{4u}\varepsilon^*(u)\,|\varphi_1(u)|.$$

Proof. For $k = 0$: $|\varphi_n|/|\varphi_1| = (|g_n|/|g_1|)E_n/E_1 \le 2n^4 e^{-\pi(n^2-1)e^{2u}}$ (Proposition 5).

For $k = 1$: using the explicit formula $\varphi_n' = g_n' E_n + g_n E_n'$, the dominant factor in E_n' is $2\pi n^2 e^{2u}$, so $|\varphi_n'|/|\varphi_1| \le (10 + 2\pi \cdot 2n^2 e^{2u}) \cdot 2n^4 e^{-\pi(n^2-1)e^{2u}}$. For $u \ge 3$, $10 \le \pi n^2 e^{2u}$, giving $|\varphi_n'|/|\varphi_1| \le 8\pi n^6 e^{2u} e^{-\pi(n^2-1)e^{2u}}$. Summing over $n \ge 2$: $|R'|/|\varphi_1| \le 8\pi e^{2u} \sum_{n\ge2} n^6 e^{-\pi(n^2-1)e^{2u}} \le 16\pi e^{2u}\varepsilon^*(u)$ (by Lemma 23).

For $k = 2$: from $E_n'' = (-4\pi n^2 e^{2u} + 4\pi^2 n^4 e^{4u})E_n$, the dominant factor is $4\pi^2 n^4 e^{4u}$, giving $|\varphi_n''|/|\varphi_1| \le 8\pi^2 n^8 e^{4u} e^{-\pi(n^2-1)e^{2u}}$. Summing: $|R''|/|\varphi_1| \le 8\pi^2 e^{4u} \sum_{n\ge2} n^8 e^{-\pi(n^2-1)e^{2u}} \le 128\pi^2 e^{4u}\varepsilon^*(u)$ (by Lemma 23). □

Lemma 25 (ΔQ far-tail bound)**.** *For $u \ge 3$:*

$$\frac{|\Delta Q(u)|}{\varphi_1(u)^2} \le 2500 e^{4u}\varepsilon^*(u).$$

Proof. From Lemma 17: $|\Delta Q|/\varphi_1^2 \le |R''|/\varphi_1 + (|\varphi_1''|/\varphi_1)(|R|/\varphi_1) + 2(|\varphi_1'|/\varphi_1)(|R'|/\varphi_1) + (|R''||R| + (R')^2)/\varphi_1^2$.

Bound on $|\varphi_1'/\varphi_1|$. Since

$$\log\varphi_1 = \log\pi + \tfrac{5}{2}u + \log h(u) - \pi e^{2u},$$

we have

$$\frac{\varphi_1'}{\varphi_1} = \tfrac{5}{2} + \frac{h'}{h} - 2\pi e^{2u}.$$

For $u \ge 3$, $h = 2\pi e^{2u} - 3$, so

$$0 < \frac{h'}{h} = \frac{4\pi e^{2u}}{2\pi e^{2u} - 3} < 3.$$

Therefore

$$\left|\frac{\varphi_1'}{\varphi_1}\right| \le \tfrac{5}{2} + 3 + 2\pi e^{2u} < 3\pi e^{2u}$$

for $u \ge 3$.

Bound on $|\varphi_1''/\varphi_1|$. Using

$$\frac{\varphi_1''}{\varphi_1} = (\log\varphi_1)'' + \left(\frac{\varphi_1'}{\varphi_1}\right)^2 = W_1 + \left(\frac{\varphi_1'}{\varphi_1}\right)^2,$$

and $|W_1| \le 5\pi e^{2u}$ for $u \ge 3$, we get

$$\left|\frac{\varphi_1''}{\varphi_1}\right| \le 5\pi e^{2u} + 9\pi^2 e^{4u} \le 10\pi^2 e^{4u}.$$

Linear terms. Using Lemma 24:

$$\begin{aligned} |R''|/\varphi_1 &\le 128\pi^2 e^{4u}\varepsilon^*, \\ (|\varphi_1''|/\varphi_1)(|R|/\varphi_1) &\le 10\pi^2 e^{4u}\varepsilon^*, \\ 2(|\varphi_1'|/\varphi_1)(|R'|/\varphi_1) &\le 2\cdot 3\pi e^{2u}\cdot 16\pi e^{2u}\varepsilon^* = 96\pi^2 e^{4u}\varepsilon^*. \end{aligned}$$

Quadratic terms. From Lemma 24,

$$\frac{|R''||R|}{\varphi_1^2} \le 128\pi^2 e^{4u}\varepsilon^{*2},$$

and

$$\frac{(R')^2}{\varphi_1^2} \le (16\pi e^{2u}\varepsilon^*)^2 = 256\pi^2 e^{4u}\varepsilon^{*2}.$$

Thus

$$\frac{|R''||R| + (R')^2}{\varphi_1^2} \le 384\pi^2 e^{4u}\varepsilon^{*2}.$$

Since $384\pi^2\cdot 10^{-1636} < 190$, we have for $u \ge 3$

$$384\pi^2 e^{4u}\varepsilon^{*2} \le 384\pi^2\cdot 10^{-1636}\cdot e^{4u}\varepsilon^* < 190\, e^{4u}\varepsilon^*.$$

Therefore the quadratic terms are absorbed into the gap $2500 - 2310 = 190$.

Conclusion. Hence

$$\frac{|\Delta Q(u)|}{\varphi_1(u)^2} \le 2500\, e^{4u}\varepsilon^*(u)$$

for all $u \ge 3$. □

Lemma 26 (W_{tail} far-tail bound)**.** *For $u \ge 3$:*

$$|W_{\text{tail}}(u)| \le 2600\, e^{4u}\varepsilon^*(u).$$

Proof. From Lemma 20: $|W_{\text{tail}}| \le |\Delta Q|/\varphi_1^2 + 3\varepsilon^*|W_1|$. Lemma 25 gives $|\Delta Q|/\varphi_1^2 \le 2500 e^{4u}\varepsilon^*$. For the second term: $|W_1(u)| \le 5\pi e^{2u}$ (Theorem 2), so $3\varepsilon^*|W_1| \le 15\pi e^{2u}\varepsilon^*$. For $u \ge 3$: $15\pi e^{2u} \le 15\pi e^{-2u}\cdot e^{4u} < e^{4u}$ (since $15\pi e^{-2u} \le 15\pi e^{-6} < 1$ for $u \ge 3$). Hence $|W_{\text{tail}}| \le 2500 e^{4u}\varepsilon^* + e^{4u}\varepsilon^* = 2501 e^{4u}\varepsilon^* < 2600\, e^{4u}\varepsilon^*$. □

Remark. Since $\varepsilon^*(3) < 10^{-1636}$ and $2600e^{12} < 10^7$, we have $2600e^{12}\varepsilon^*(3) < 10^{-1629} \ll 1$. Hence the bound in Lemma 26 is overwhelmingly small at $u = 3$.

Lemma 27 (Monotonicity of W_1 for $u \geq 3$)**.** *Let*

$$h(u) = 2\pi e^{2u} - 3, \qquad W_1(u) = -\frac{24\pi e^{2u}}{h(u)^2} - 4\pi e^{2u}.$$

Then $W_1'(u) < 0$ for all $u \geq 3$.

Proof. Differentiate:

$$W_1'(u) = -8\pi e^{2u} - 48\pi e^{2u}h^{-2} + 48\pi e^{2u}h^{-3}h'.$$

Since $h' = 4\pi e^{2u} = 2(h+3)$, this becomes

$$W_1'(u) = -8\pi e^{2u} - 48\pi e^{2u}h^{-2} + 96\pi e^{2u}(h+3)h^{-3}.$$

Factoring $-8\pi e^{2u}h^{-3}$, we obtain

$$W_1'(u) = -\frac{8\pi e^{2u}}{h^3}\Big(h^3 + 6h - 12(h+3)\Big) = -\frac{8\pi e^{2u}}{h^3}\Big(h^3 - 6h - 36\Big).$$

For $u \geq 3$, $h(u) = 2\pi e^{2u} - 3 > 3.92$, and

$$h^3 - 6h - 36 > 0.$$

Therefore $W_1'(u) < 0$ for all $u \geq 3$. □

Lemma 28 (Monotonicity of far-tail expression)**.** *The function $u \mapsto e^{4u}\varepsilon^*(u)$ is strictly decreasing on $[3,\infty)$. Consequently $W_1(u) + 2600\, e^{4u}\varepsilon^*(u)$ is strictly decreasing on $[3,\infty)$.*

Proof. Since

$$\varepsilon^*(u) = 2\sum_{n\geq 2} n^4 e^{-\pi(n^2-1)e^{2u}},$$

differentiation gives

$$\varepsilon^{*\prime}(u) = -4\pi e^{2u}\sum_{n\geq 2} n^4(n^2-1)e^{-\pi(n^2-1)e^{2u}}.$$

As $n^2 - 1 \geq 3$ for $n \geq 2$,

$$\varepsilon^{*\prime}(u) \leq -12\pi e^{2u}\sum_{n\geq 2} n^4 e^{-\pi(n^2-1)e^{2u}} = -6\pi e^{2u}\varepsilon^*(u).$$

Therefore

$$\frac{d}{du}\Big(e^{4u}\varepsilon^*(u)\Big) = e^{4u}(4\varepsilon^*(u) + \varepsilon^{*\prime}(u)) \leq e^{4u}(4 - 6\pi e^{2u})\varepsilon^*(u) < 0$$

for $u \geq 3$. This proves $e^{4u}\varepsilon^*$ is strictly decreasing. Strict monotonicity of W_1 follows from Lemma 27. □

E.7 Complete proof of Lemma 9

Proof of Lemma 9: uniform tail bound. The proof uses a single route.

Step 1. From Lemma 20: $|W_{\text{tail}}(u)| \leq B(u)$ where $B(u) = |\Delta Q(u)|/\varphi_1(u)^2 + 3\varepsilon^*(u)|W_1(u)|$.

Step 2. Lemma 21 certifies by interval arithmetic that $B(u) \leq 1.12 \times 10^{-23}$ for all $u \in [1, 3]$ (maximum at $u = 1$). In particular $|W_{\text{tail}}(u)| \leq B(u) \leq 1.12 \times 10^{-23}$ for all $u \in [1, 3]$.

Step 3. Lemma 21 further certifies upper$(W_1(I_i) + B(I_i)) < 0$ on each of the 101 intervals covering $[0.99, 3.01] \supset [1, 3]$. Since $W = W_1 + W_{\text{tail}}$ and $|W_{\text{tail}}| \leq B$:

$$W(u) = W_1(u) + W_{\text{tail}}(u) \leq W_1(u) + B(u) < 0,$$

with minimum margin 91.3 at $u = 1.0$. □